

$$\begin{array}{l} R \\ L^2 \\ f: \\ [0,T] \rightarrow \\ R \end{array}$$

$$\sum_{j=0}^{n-1} f(s_j)(t_{j+1}-t_j),$$

$$\begin{array}{l} (1) \\ 0= \\ t_0< \\ t_1< \\ \vdots< \\ t^n= \\ T \\ s_j \\ [t_j,t_{j+1}] \\ j \\ s_j\in \\ [t_j,t_{j+1}) \end{array}$$

$$\sum_{j=0}^{n-1} f(s_j)(W(t_{j+1})-W(t_j)).$$

$$\begin{array}{l} (2) \\ s_j \\ [t_j,t_{j+1}] \\ f(t)= \\ W(t) \\ 0= \\ t_0^n< \\ t_1^n< \\ \vdots^n< \\ t^n= \\ T^n= \\ t_j^n \\ jT \\ [0,T] \\ n^2 \\ L^2 \end{array}$$

$$\lim_{n\rightarrow\infty}\sum_{j=0}^{n-1}W(t_j^n)\left(W(t_{j+1}^n)-W(t_j^n)\right)$$

$$(3)$$

$$\lim_{n\rightarrow\infty}\sum_{j=0}^{n-1}W\left(t_{j+1}^n\right)\left(W\left(t_{j+1}^n\right)-W\left(t_j^n\right)\right).$$

$$(4)$$

$$a(b-a)=\frac{1}{2}(b^2-a^2)-\frac{1}{2}(a-b)^2,$$

$$(5)$$

$$b(b-a)=\frac{1}{2}(b^2-a^2)+\frac{1}{2}(a-b)^2$$

$$(6)$$

$$\begin{array}{l} a= \\ W(t_j^n) \\ b= \\ W(t_{j+1}^n) \\ \sum_{j=0}^{n-1} W(t_j^n)(W(t_{j+1}^n)- \\ W(t_j^n))= \\ \frac{1}{2}\sum_{j=0}^{n-1}(W(t_{j+1}^n)^2- \\ W(t_j^n)^2)- \\ \frac{1}{2}\sum_{j=0}^{n-1}(W(t_{j+1}^n)- \\ W(t_j^n))^2 \\ = \\ \frac{1}{2}(W(T)^2- \\ W(0)^2)- \\ \frac{1}{2}\sum_{j=0}^{n-1}(W(t_{j+1}^n)- \\ W(t_j^n))^2 \\ = \\ \frac{1}{2}W(T)^2- \\ \frac{1}{2}\sum_{j=0}^{n-1}(W(t_{j+1}^n)- \\ W(t_j^n))^2. \\ [0,T] \end{array}$$

$$\frac{1}{2}W(T)^2+\frac{1}{2}T.$$

$$s_j^{[t_j,t_{j+1}]}$$

$$\mathcal{F}_{s_j^t}^t=$$

$$t_j$$

$$\mathcal{F}_t$$

$$f(t),t\geq 0$$

$$\textbf{random step process}$$

$$0=t_0< t_1< \cdots < t_n$$

$$\eta_0,\eta_1,\ldots,\eta_{n-1}$$

$$f(t)=\sum_{j=0}^{n-1}\eta_j\mathbf{1}_{[t_j,t_{j+1})}(t),$$

$$(7) \quad \begin{array}{l} \eta_j \\ \mathcal{F}_{t_j} \\ \dot{=} \\ 0,1,\ldots,n-1 \\ M_{\text{step}}^2 \\ \eta_j \\ \mathcal{F}_{t_j} \\ f(t) \\ \mathcal{F}_t \\ \eta_j \\ f(t) \\ t \\ M_{\text{step}}^2 \\ af+bg\in M_{\text{step}}^2 \\ f,g\in M_{\text{step}}^2 \\ a,b\in R \\ f\in M_{\text{step}}^2 \end{array}$$

$$I(f)=\sum_{j=0}^{n-1}\eta_j\left(W(t_{j+1})-W(t_j)\right).$$

$$(8) \quad \begin{array}{l} f\in M_{\text{step}}^2 \\ I(f) \\ I(f)\in L^2 \end{array}$$

$$E\left(|I(f)|^2\right)=E\left(\int_0^\infty |f(t)|^2\,dt\right).$$

$$(9) \quad \begin{array}{l} W(t_{j+1})-W(t_j) \\ \Delta_j W \\ t_{j+1}-t_j \\ \Delta_j t \end{array}$$

$$(10) \quad E(\Delta_j W)=0 and E\left((\Delta_j W)^2\right)=\Delta_j t.$$

$$(11) \quad |I(f)|^2=\sum_{j=0}^{n-1}\sum_{k=0}^{n-1}\eta_j\eta_k\Delta_jW\Delta_kW=\sum_{j=0}^{n-1}\eta_j^2(\Delta_jW)^2+2\sum_{k<j}\eta_j\eta_k\Delta_jW\Delta_kW.$$

$$\eta_j\Delta_jW$$

$$E\left(n^2(\Delta_n W)^2\right)=E\left(n^2\right)E\left((\Delta_n W)^2\right)=E\left(n^2\right)\Delta_n t$$

$$\int_0^T W(t) dt.$$

$$f, g \in$$

$$M_t^2$$

$$\alpha, \beta \in$$

$$R \leq$$

$$0 \leq$$

$$\frac{s}{t} \geq$$

Linearity:

$$(54) \quad \int_0^t (\alpha f(r) + \beta g(r)) dW(r) = \alpha \int_0^t f(r) dW(r) + \beta \int_0^t g(r) dW(r);$$

Isometry:

$$(55) \quad E \left(\left| \int_0^t f(r) dW(r) \right|^2 \right) = E \left(\int_0^t |f(r)|^2 dr \right);$$

**Martingale
property:**

$$(56) \quad E \left(\int_0^t f(r) dW(r) \mid \mathcal{F}_s \right) = \int_0^s f(r) dW(r).$$

$$f$$

$$g$$

$$M_t^2$$

$$\mathbf{1}_{[0,t)} f$$

$$\mathbf{1}_{[0,t)} g$$

$$M_{\text{step}}^2$$

$$f_1, f_2, \dots$$

$$g_1, g_2, \dots$$

$$M_{\text{step}}^2$$

$$\mathbf{1}_{[0,t)} f$$

$$\mathbf{1}_{[0,t)} g$$

$$\mathbf{1}_{[0,t)} (\alpha f +$$

$$\beta g)$$

$$\alpha f_1 +$$

$$\beta g_1, \alpha f_2 +$$

$$\beta g_2, \dots$$

$$??$$

$$(57) \quad I(\alpha f_n + \beta g_n) = \alpha I(f_n) + \beta I(g_n)$$

$$\frac{n}{\infty} \rightarrow$$

$$(58) \quad I(\mathbf{1}_{[0,t)}(\alpha f + \beta g)) = \alpha I(\mathbf{1}_{[0,t)} f) + \beta I(\mathbf{1}_{[0,t)} g),$$

$$\mathbf{1}_{[0,t)} f$$

$$M_{\text{step}}^2$$

$$??$$

$$f$$

$$M_t^2$$

$$\mathbf{1}_{[0,t)} f$$

$$M_{\text{step}}^2$$

$$f_1, f_2, \dots$$

$$M_{\text{step}}^2$$

$$\mathbf{1}_{[0,t)} f$$

$$??$$

$$(59) \quad E(I(\mathbf{1}_{[0,t)} f_n) \mid \mathcal{F}_s) = I(\mathbf{1}_{[0,s)} f_n)$$

$$\frac{n}{\infty} \rightarrow$$

$$(60) \quad E(I(\mathbf{1}_{[0,t)} f) \mid \mathcal{F}_s) = I(\mathbf{1}_{[0,s)} f),$$

$$\mathbf{1}_{[0,s)} f_1, \mathbf{1}_{[0,s)} f_2, \dots$$

$$M_{\text{step}}^2$$

$$\mathbf{1}_{[0,s)} f$$

$$(61) \quad I(\mathbf{1}_{[0,s)} f_n) \rightarrow I(\mathbf{1}_{[0,s)} f) \text{ in } L^2 \text{ as } n \rightarrow \infty.$$

$$\mathbf{1}_{[0,t)} f_1, \mathbf{1}_{[0,t)} f_2, \dots$$

$$M_{\text{step}}^2$$